EXP NO: 11 Routing at network layer

a) Aim : To configure static routing between two network using cisco packet ~~tracer~~

Procedure : 1) Place two router, two switches and four pcs.

*) connect pcs to switch and switch to router.

*) connect the two router using an ethernet link.

*) Assign ip address to all pcs and router interface

*) configure static router on both router.

Router 1 configuration;

```
enable
configure terminal
interface gigabit ethernet 0/0
ip address 192.168.1.1.255.255.0
no shutdown
exit
```

Router 2 configuration;

```
enable
configure terminal
interface gigabit ethernet 0/0
ip address 192.168.2.1 255.255.255
no shutdown
exit
```

Result; Statics routing is configured successfully.

b) Simulate RIP using cisco packet tracer

Aim; To configure the routing Information protocol (RIP) for communication between different network.

Procedure; *) create a topology with two router and two LANs

*) Assign IP addresses to router interface and PCs.

*) Enable RIP on both router

*) Advertise the connected network

*) Test connectivity using the ping command.

Router 1 configuration;

```
enable
configure terminal
router rip
 version 2
no auto - summary
network 192.168.1.0
 network 10.0.0.0
 end
```

Router 2 configuration;

```
enable
configure terminal
router rip
 version 2
no auto - summary
network 192.168.20
 network 10.0.00
end
```

testing ; show ip route

ping 192.168.2.2

Result ; RIP is configured successfully and the router exchange routing information to communicate between network.